

SECTION 33 32 11.10
PUMP CONTROL SYSTEMS (EMEA)

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PART 0 PROJECT-SPECIFIC AMENDMENTS

The following amendments to this Section apply only to this Project:

- 1.2.B: Replace '[XX]' with '14'

PART 1 GENERAL

1.1 SCOPE

- Provide a complete Pump Control System for wastewater pump stations by monitoring liquid level in wet well using pressure transducer-type level sensors.
- The Pump Control Panel with all associated appurtenances constitutes the Pump Controls System. Refer to Drawings for additional information and materials list.
- The major Pump Control Panel components include: Programmable logic level controller/Remote Terminal Units, direct drive motor drive controllers, shorting (bypass) and isolation contactors, protective relays, intrinsically safe relay, control relays, pushbuttons, terminals, circuit breakers, fuses, surge protective device, enclosure, wiring, wire-ways, and all other necessary appurtenances and associated materials lists as indicated on drawings.
- Major components required in the complete pump control system are specified in this specification. Other components are not directly specified but are nevertheless required and shall be provided by the pump control panel fabricator. The panel fabricator shall provide a completely operational, prewired, pretested pump control system.
- The Pump Control System supplier shall purchase and install new, good quality materials to provide a completely functioning ATEX certified system to (ATEX EU directive 94/9/EC).
 - CE marking. Any materials in the material list that could possibly jeopardize the panel's CE certification shall be substituted by the Pump Control Panel supplier/fabricator with ATEX certified.
 - CE marking products. The Contractor is responsible to immediately notify the Engineer of any and all changes to the materials list as a result of this requirement. Detailed justification shall be provided for each substituted part.

1.2 SUBMITTALS

- Section 01 33 00 - Submittal Procedures.
- Shop Drawings: indicate control panel layouts, wiring connections and diagrams, dimensions, and support points. The layout shall exactly match the projects' control panel drawing. Only Engineer-approved, justified changes are allowed and written approval is required prior to deviation from the projects control panel drawings. The Engineer will require ~~XX~~14 calendar days to review design deviations. Schedule for submittal review is the Contractor's responsibility.
- The Pump Controls System supplier/fabricator shall provide for the project and to the Engineer a completely record set of drawings in ACAD format and in PDF format. Complete materials list, with manufacturer and part numbers shall be included. Drawings and Materials Lists shall be provided in electronic format; three (3) copies are required.

- D. Product Data: Submit catalog data for each component showing electrical characteristics, physical dimensions and connection requirements. Product Data for components may be provided as PDF catalog cut-sheets electronically, or neatly bound and tabbed in 3-ring binders with table of contents. All component selections shall be highlighted or otherwise outlined to indicate to the Engineer and Owner which components are selected. If proper marking has not been done, the submittal will be returned to the Contractor without comments and will be unequivocally rejected.
- E. Materials used shall be as per the attached material lists at the back of this specification section. "Or-equal" substituted components shall be noted in the panel fabricator's submittal with complete information required in paragraph C above.

1.3 QUALIFICATIONS

- A. Panel Fabricator: Company specializing in manufacturing products specified in this Section with minimum five years of experience.

PART 2 PRODUCTS

2.1 PUMP CONTROL SYSTEM OVERVIEW

- A. System Requirements: The controller used for this project shall be an advanced microprocessor based pump controller provided by the equipment supplier that is compatible with the pump manufacturer's standard equipment. Standardization of pump control panel design and interchangeability of parts necessitate this fact. The equipment supplier shall have unit responsibility for proper integration, compatibility and proper function of pumps and controls for this pump station.
- B. The Pump Control Panel shall be provided with a main circuit breaker with handle operator. All circuit breaker handle operators shall properly fit respective circuit breakers and shall not impede swing-out door operation. Improperly fitted circuit breaker extension handles, or any problems caused by the circuit breaker handles shall be repaired by the Contractor at his expense and at no additional cost to Owner.
- C. Motor protection:
 - 1. The pump controller shall be used to provide advanced motor protection and data analysis of the system performance as well as integration with SCADA system.
 - 2. For panels serving submersible pumps, each pump motor shall be wired to an electrode relay providing high probe potential for the leak detection system built into the oil filled motor shaft housing. The electrodes will activate if moisture content in the oil sealed area reaches abnormal levels. Electrode activation shall open the control circuit to the pump motor starter preventing possible damage to the pump motor from water ingestion. Pump will be placed out of service until maintenance is performed on the unit and control system reset.
- D. Enclosure: See Section 2.8 in this specification. All required equipment shall be installed inside a stainless steel enclosure equipped with a subpanel and interior swing open dead front door. The enclosure is located in a non-hazardous but corrosive location, outdoors exposed to weather. Circuits originating from the enclosure shall be suitable for the hazardous area and comply with IEC 60079. Separate all wiring for intrinsically safe circuitry in accordance with IEC 60079-11.

- E. System Voltage Protection: Transient Voltage Surge Suppressor (TVSS) or surge protective device shall be provided, pre-wired inside the enclosure. The IEC 61643-1 and EU standard EN 61643-11, surge protective device (SPD) shall provide 4-mode, line-voltage surge protection for the control panel incoming power supply. The SPD shall have a minimum surge rating of 200 kA/phase, with each phase having its own status indicator. Surge current modes are Line to Neutral, Line to Ground, Neutral to Ground, and Line to Line, and each mode shall have a minimum of 100 kA/phase protection. The TVSS shall be provided with a line-side, manual disconnecting switch inside the panel to simultaneously disconnect all ungrounded conductors for maintenance. Lead lengths shall be as short as possible.
- F. Provide two (2) 20A, 1-pole, circuit breakers inside the Pump Control Panel per material list and drawings. Circuit breakers shall be bolt-on or din-rail mounted, 10kAIC at 120/240 Volts AC rated interrupting capacity, minimum.
- G. Backup power supplies: Provide a 12 Volt DC supply inside Pump Control Panel for the SCADA Radio system. The system shall accept 240 V ac input voltage and consist of a DC power supply/charger, and battery:
 - 1. The 12 Volt DC power supply/charger shall be 55 Watts, minimum, with automatic reset for over currents. Output current 0-4 Amps; ± 0.5 percent load regulation, ± 1 percent output voltage is nominal 13.8 V dc with regulation 100 mV Peak-peak ripple. The charger shall be integral with power supply.
 - 2. The 12 Volt DC rechargeable battery shall be SLA, 12 Ah, with nut and bolt, or spade lug terminations.
- H. Remote Alarm and Points Reporting - The Pump Control Panel shall report alarms and points remotely through the SCADA system. Alarms and points shall be reported through SCADA for any of the following reasons:
 - 1. High water level in wetwell.
 - 2. Low-Low water level in the wetwell.
 - 3. The moisture detection system indicates moisture in the oil chamber between the motor seals (for each pump).
 - 4. The motor winding temp switches indicate high motor temperature caused by improper motor cooling or motor overload (for each pump).
 - 5. Overload protective devices in motor controls circuit indicate motor overload (for each pump).
 - 6. Electrical failure detected by the power failure control relay, or by the phase monitor.
 - 7. Motor branch circuit breaker disconnecting device left in "off" position.
 - 8. Pump Run Time (for each pump).
 - 9. Discrete Hand-Off-Auto Status for each pump (In Auto or Not In Auto).
 - 10. Wetwell Water Level (Analog).
 - 11. Flow Meter Indication (Totalized Flow and Rate of Flow)
 - 12. Other alarm events and points as shown on the pump control drawings.
- I. Wiring:
 - 1. Control wire shall be MTW, 90 degree C, stranded copper 1.5 mm² for 24 V dc power wiring and 2.5 mm² for 230 V ac power wiring as a min. size with conductors sized appropriate to current required. This is consistent with Section 23 09 10 - Controls-Critical-BMS. Signal wiring may be 0.75 mm² as a min. size with conductors sized appropriate to current required. This is consistent with Section 23 09 10 - Controls-Critical-BMS.

2. All wiring should be neatly grouped in plastic wire trough except wiring from the backplate to the door shall be done in separate bundled harnesses.
3. All wires shall have a wraparound wire identification number as shown in the wiring diagram at both ends.
4. All components shall be identified with the same number as shown in the control panel schematic drawings.
5. Control wiring terminations shall be performed using properly sized, Thomas & Betts, Sta-kon type insulated ferrules, or ring terminals.
6. Wiring shall be secured using stainless steel tie wraps or cable clamps. Plastic or nylon wire ties are not acceptable.

2.2 CONTROL SWITCHES AND STATIONS

- A. Manufacturers:
 1. ABB.
 2. Rockwell Automation/Allen-Bradley.
 3. Eaton/Cutler-Hammer.
 4. Square D.
 5. Substitutions: Not Permitted.
- B. Product Description: IEC 60947-5-1 certified, heavy-duty, corrosion resistant devices.
- C. Contacts: Form Z, or as required by control panel drawings.
- D. Contact Ratings: Class A600.
- E. Pushbutton Operator: Non-illuminated, momentary contact, flush type. Use push buttons listed as suitable for outdoors, weather-exposed applications.

2.3 MAGNETIC CONTROL RELAYS

- A. Manufacturers:
 1. ABB.
 2. Rockwell Automation/Allen-Bradley.
 3. Eaton/Cutler-Hammer.
 4. Square D.
 5. G.E.
 6. Substitutions: Not Permitted.
- B. Product Description: IEC 60947-5-1 certified, Class A300 magnetic control relay. Relays shall have industry standard wiring and pin and base arrangements. Control relay coils shall be equipped with appropriate voltage-transient protection.
- C. Contacts: Form C, or as selected by panel fabricator.
- D. Contact Ratings: Class A150, minimum.
- E. Coil Voltage: 230 V ac/110 V ac, 50 Hz or 24 V dc, or as required on Drawings.
- F. Enclosure: Ingress protection (IP), type to meet conditions of installation.

2.4 SOLID-STATE CONTROL RELAYS

- A. Manufacturers:
 - 1. ABB.
 - 2. Rockwell Automation/Allen-Bradley.
 - 3. Eaton/Cutler-Hammer.
 - 4. Square D.
 - 5. G.E.
 - 6. Substitutions: Not Permitted.
- B. Product Description: IEC 60947-5-1 certified, solid-state electronic relay.
- C. Contacts: Form C. or as selected by panel fabricator.
- D. Contact Ratings: Class A150, minimum.
- E. Coil Voltage: 230 V ac/100 V ac, 50 Hz or 24 V dc. AC, or 12 Volts DC, or as required on Drawings.
- F. Listing: CE certified.

2.5 TIME DELAY RELAYS

- A. Manufacturers:
 - 1. ABB.
 - 2. Artisan.
 - 3. Rockwell Automation/Allen-Bradley. Design Basis is #700-RTC20200U1 with adjustable off-timer delay.
 - 4. Eaton/Cutler-Hammer.
 - 5. Square D.
 - 6. Idec.
 - 7. Substitutions: Not Permitted.
- B. Product Description: IEC 60947-5-1 certified, solid state, sealed switch timing relay. Unit is an initiated, switch controlled, delay-on-break timer operating a set of contacts. Select optional timer settings as indicated on drawings.
- C. Contacts: NEMA B600, 600 V ac, 5 Amps.
- D. Coil Voltage: 230 V ac/110 V ac, 50 Hz or 24 Vdc, as required on Drawings.
- E. Listing: CE certified

2.6 LEVEL SENSOR

- A. Manufacturers:
 - 1. Flygt
- B. Model Number: LTU 601

2.7 CONTROL POWER TRANSFORMER

- A. Manufacturers:

1. ABB.
2. Rockwell Automation/Allen-Bradley.
3. Eaton/Cutler-Hammer.
4. Square D.
5. G.E.
6. Substitutions: Not Permitted.

B. Product Description: EN 61558-2, machine tool transformer with isolated secondary winding. CE certified.

C. Power Rating: 5 kVA, or as rated otherwise on the Drawings.

D. Voltage Rating: 416/240 V primary, 110 Volts or 24 V dc secondary.

2.8 CONTROL PANEL ENCLOSURES

A. Manufacturers:

1. Rittal cabinets
2. Saginaw
3. B-Line
4. Nvent Hoffman. – Design basis Enclosure and Air Conditioning unit
5. EMF
6. Substitutions: as agreed via Technical Submission

B. Product Description: Outside cabinets are IP 65 as a minimum. 304 or 316 stainless steel – no exceptions. The IP rating to EN 60529 rating may be reduced to IP54 for indoor panels based upon installation of the Air Conditioner unit and the 3-point, 90 degree turn lockable latch kits that are required on this panel. Panel shall have a hinged, interior door to provide mounting surface for controller displays, labels, and other required components.

C. Pump Control Panel Minimum Size: All pump control panels over 30 HP shall be free-standing, with the panel firmly secured to the finished floor. The minimum allowed free-standing enclosure size shall be 74 by 60 by 24 inches.

D. Triplex stations may be equipped with two (2) side by side free-standing enclosures; however, both enclosures will require separate air conditioners. Accessory feet kits shall be provided with adequate clearance from the finished floor such that sufficient space is included to install conduit stub-ups from underground and their associated sealing fittings and hardware and hubs.

E. Shop-fabricate the Pump Control Panel to IP 54 as a minimum.

F. Box Size: as required to meet project requirements. Ensure adequate wiring space is provided, and thermal/air flow needs of various components are considered in sizing of enclosure.

G. Fronts: 304 or 316 stainless steel, fully gasketed, dead-front, surface mounted type with 3-point, quarter-turn latching, lockable door handle with concealed hinge. Hinges shall be completely rust-free under all circumstances. Enclosures, hinges or hardware that rusts, including "surface" rust where pitting is evident, shall cause the entire panel to be replaced and any and all associated repair work and materials shall be at the Contractor's expense and at no additional costs to the project.

- H. Finish: Manufacturer's standard.
- I. Knockouts: Performed by field Contractor.
- J. Furnish grounded metallic or other acceptable barriers to form separate compartments for wiring of different systems and voltages. Isolate Intrinsically Safe equipment according to IEC 60079-11.

2.9 TERMINAL BLOCKS

- A. Manufacturers:
 - 1. ABB.
 - 2. Weidmuller.
 - 3. Eaton/Cutler-Hammer.
 - 4. Square D.
 - 5. G.E.
 - 6. Substitutions: per Section 01 33 23.
- B. Product Description: IP 2X, terminal blocks.
- C. Power Terminals: Unit construction type with closed back and tubular pressure screw connectors, rated 600 Volts. Suitable for use with copper wire.
- D. Signal and Control Terminals: Modular construction type, suitable for channel mounting, with tubular pressure screw connectors, rated 600 Volts. Suitable for use with copper wire.
- E. Include ground terminal strips. All panel sections shall be individually earthed including but not limited to roof section, side panels, gland plates and door.
- F. Terminals shall be properly secured to sub-panel or din rails. If conductors are causing terminals to become stressed by over tension, or improper bending space or any other reasons causing the terminals to bend, deform, or become overburdened, the Contractor shall make repairs at his expense, at no additional cost to the project.

2.10 PLASTIC TRUNKING

- A. Manufacturers:
 - 1. ABB.
 - 2. Panduit.
 - 3. Nvent Hoffman.
 - 4. Thomas & Betts.
 - 5. Substitutions: Permitted as required by panel fabricator.
- B. Product Description: Low smoke, zero halogen plastic trunking or snap-on cover.

2.11 SCADA COMMUNICATIONS EQUIPMENT

- A. Required Equipment:
 - 1. Radio. Microwave Data Systems (MDS) 4710 or as required by the Owner.
 - 2. Antenna. Cushcraft PC457N for 450 MHz frequency operation or as required by the Owner.

3. Coaxial Cable. Andrew Heliax LDF4-50A, including coaxial jumper inside panel, and all connectors.
 4. Surge suppressor. Polyphaser bulkhead fitting.
 5. Coaxial Grounding Kit and miscellaneous appurtenances.
 6. Path Study shall be performed by the SCADA Subcontractor after consultation with the Contractor. Schedule of Path Studies shall be provided to the Engineer and the Owner 30 days prior to actual study.
 7. Substitutions: Permitted as required by the Owner and SCADA equipment supplier.
- B. System Description:
1. All control signals, status signals, alarm and or process variable data generated by the RTU shall be transmitted and received between the central location and the remote site via the SCADA radio communication system. Ensure that Radio Communication of Charleston performs a path study between remote pump station locations and repeaters or master stations as required.
 2. The RTU is the MultiTrobe MultiSmart Controller with DNP3, flow calculation, logic engine, and ADAMS I/O expansion. The RTU provides signal input to the Radio. Power input to the Radio comes from the 12 VDC power supply unit that is in the Pump Control Panel described in 2.1:G above.
 3. Radio output is transmitted via a coaxial cable to the antenna. The Antenna shall be mounted by Contractor. All coaxial connections or terminations are to be made by RCC (supplier) and the entire coaxial cable line shall be tested for losses, grounds and continuity by RCC. Any damaged components, cable, or high losses found or discovered through testing will be repaired by the Contractor at no additional cost to the project.
 4. Any additional components, testing, or required equipment for a completely operational SCADA radio communications shall be provided by the Contractor at no additional costs to the project.
- C. SCADA System Assembly
1. The radio, coax surge suppressor, 6 ft. coaxial cable jumper, communications connectors (DB 25) and any other radio required hardware shall be provided to the Pump Control Panel fabricator for installation in the Panel. System shall be completely integrated.
 2. Antenna, feed-line coaxial cable, coaxial cable supporting PVC or Nylon insulated straps (to secure coaxial cable to wood pole) and the raceway (helix) cable bonding/grounding kit shall be provided by RCC to Contractor for installation.
 3. Contractor shall install an antenna, upper connector and waterproof kit, helix and grounding/bonding kit in accordance with the scope of work provided by RCC, Inc. Additionally, the wood antenna pole, all raceway and raceway supports shall be provided and installed by the Contractor.
 4. The Contractor shall ensure that RCC completes the radio installation and activates the radio after Contractor has completed the installation, and permanent power is available on site.

PART 3 EXECUTION

3.1 EXISTING WORK

- A. Where necessary, disconnect and remove abandoned controls and relays remaining at job site. Maintain station electrical service and control at all times except for switchover to new controls. Coordinate with the Owner and Engineer prior to switchover.
- B. Where existing equipment is present, dispose of equipment/materials as directed by the Owner and Engineer.
- C. The Owner may elect to salvage certain equipment. If the Owner decides to salvage any equipment, wiring or materials, it shall be returned to the Owner cleaned and in an undamaged condition. Coordinate with the Owner and Engineer for a location to store or stage salvaged materials.

3.2 INSTALLATION

- A. Install the assembled and pre-tested pump control panel on the electrical equipment rack, or bolt down to the concrete pad if free-standing, as shown on the project design drawings.
- B. Install enclosures and boxes plumb.
- C. Make electrical wiring interconnections as indicated on Drawings, and as otherwise required for a completely functioning control system.
- D. Install engraved plastic nameplates.
- E. Ground and bond control panels, raceway, and intrinsically safe barriers and relays.
- F. Cables/Conduits entering the control panel from the wet well must be sealed in accordance with ATEX certified system to (ATEX EU directive 94/9/EC) Specifications - provide sealing compound to the Owner and Engineer. Contractor will install it. Separate wiring for intrinsically safe circuits for the level probe and float switch according to IEC 60079-11.
- G. Clean the enclosure completely, vacuum out any wire strippings, clippings or metallic filings that may be found. Use manufacturer recommended cleaning Agents. Improperly cleaned cabinets are cause for rejection of construction work.

END OF SECTION

REVISION HISTORY

Version	Date	Description and location of change
0.1	August 2024	Initial version – BOD Draft submission
0.2	September 2024	BOD Final Submission

0.3	February 2025	Draft IFC Updates
1.0	12 th of March 2025	Final IFC Issue